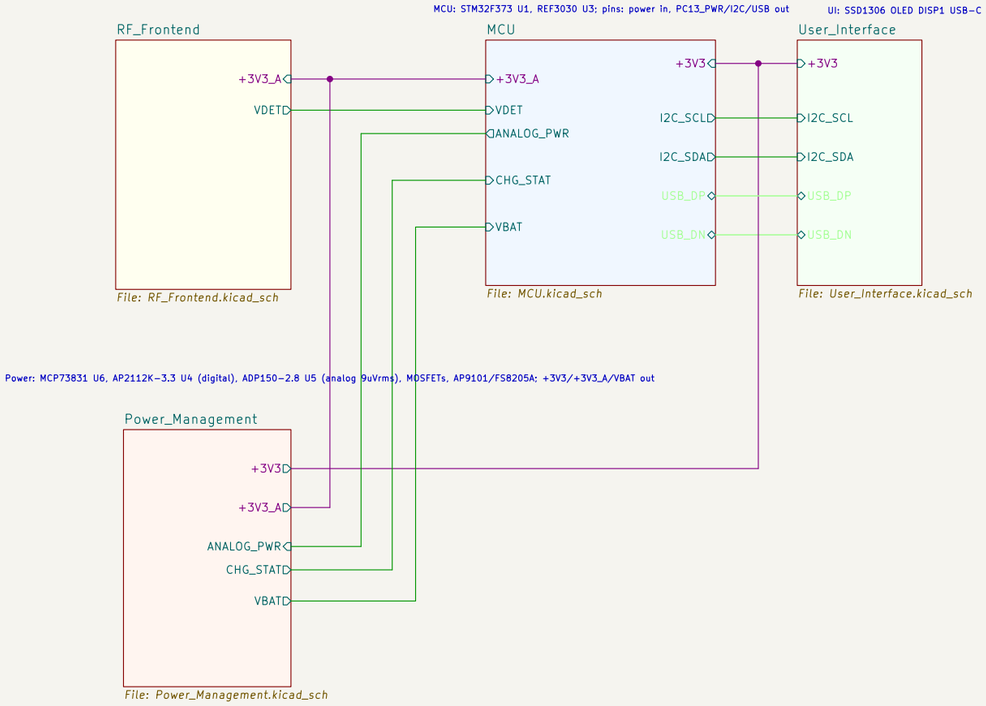
RFef to: https://www.eevblog.com/forum/rf-microwave/how-to-properly-use-these-sma-connectors/new/#new

ad8317_powermeter.kicad_sch — top-level A3: 4 hierarchical sub-sheets

RF Frontend: SMA J1, AD8317 U2, RC filter; +3V3_A in, VDET/AGND_RF out

MCU: STM32F373 U1, REF3030 U3; pins: power in, PC13_PWR/I2C/USB out

UI: SSD1306 OLED DISP1 USB-C CONN1; I2C/USB



Power: MCP73831 U6, AP2112K-3.3 U4 (digital), ADP150-2.8 U5 (analog 9uVrms), MOSFETs, AP9101/FS8205A: +3V3/+3V3_A/VBAT out

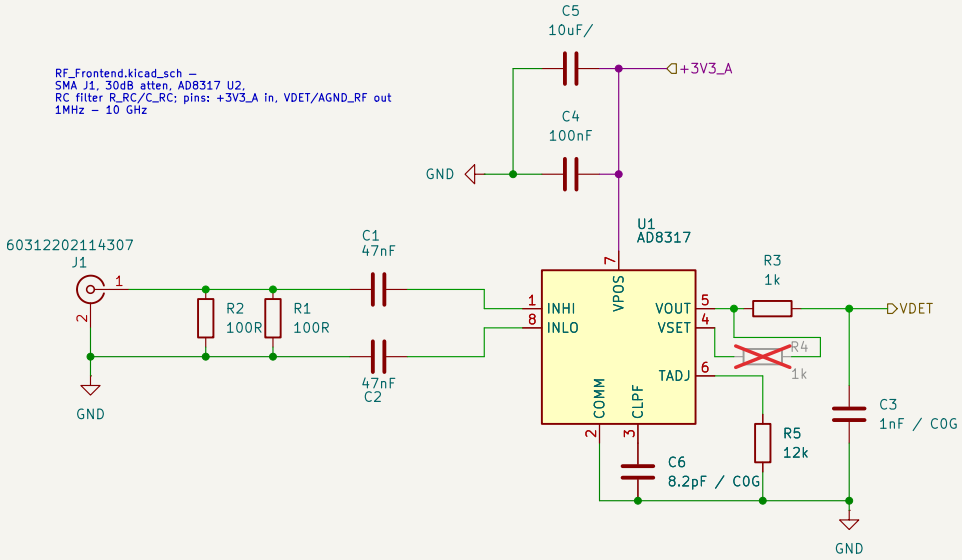
AD8317 log detector (U2), RC filter R_RC=1kR C_RC=1nF C0G
SMA (J1 Wurth Elektronik 60312202114307)
Author

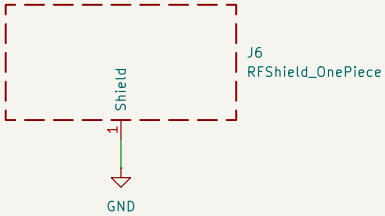
Sheet: /
File: ad8317_powermeter.kicad_sch

Title: RF Frontend

Size: A3	Date: 2026-08-07	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/5

RF_Frontend.kicad_sch —
SMA J1, 30dB atten, AD8317 U2,
RC filter R_RC/C_RC; pins: +3V3_A in, VDET/AGND_RF out
1MHz – 10 GHz





AD8317 log detector (U2), RC filter R_RC=1kR C_RC=1nF C0G
SMA (J1 Wurth Elektronik 60312202114307)

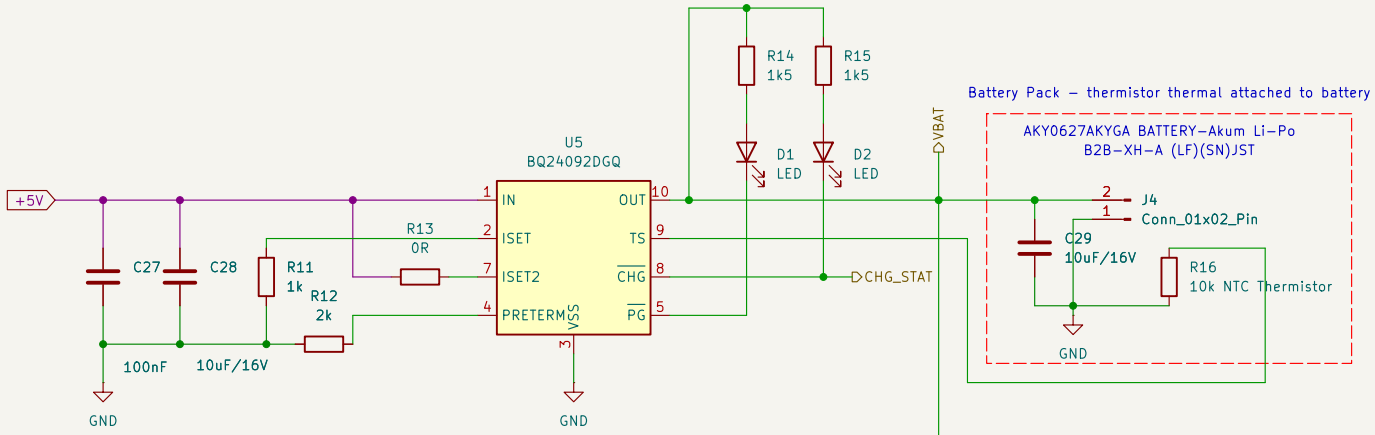
Author

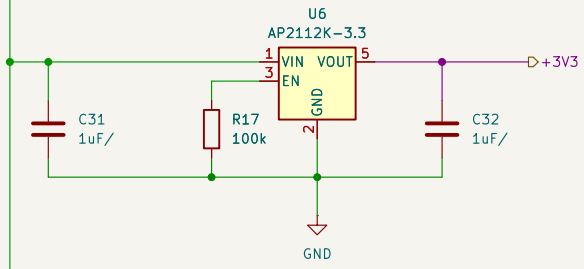
Sheet: /RF_Frontend/
File: RF_Frontend.kicad_sch

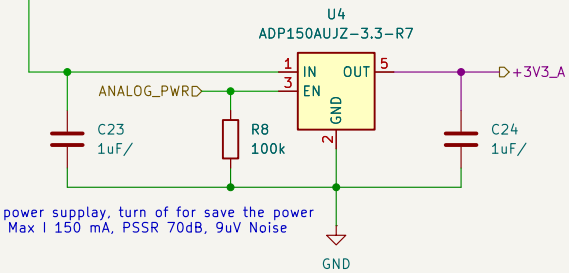
Title: RF Frontend

Size: A4	Date: 2026-07-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 2/5

Power_Management.kicad_sch — MCP73831 charger U6, AP2112K-3.3 digital LDO U4, ADP150AUJZ-2.8 analog LDO U5 (9uVrms), AP9101/FS8205A protection; pins: PC13_PWR in, +3V3/+3V3_A/VBAT/CHG_STAT out







Analog power supplay, turn of for save the power
3V3 Max I 150 mA, PSSR 70dB, 9uV Noise

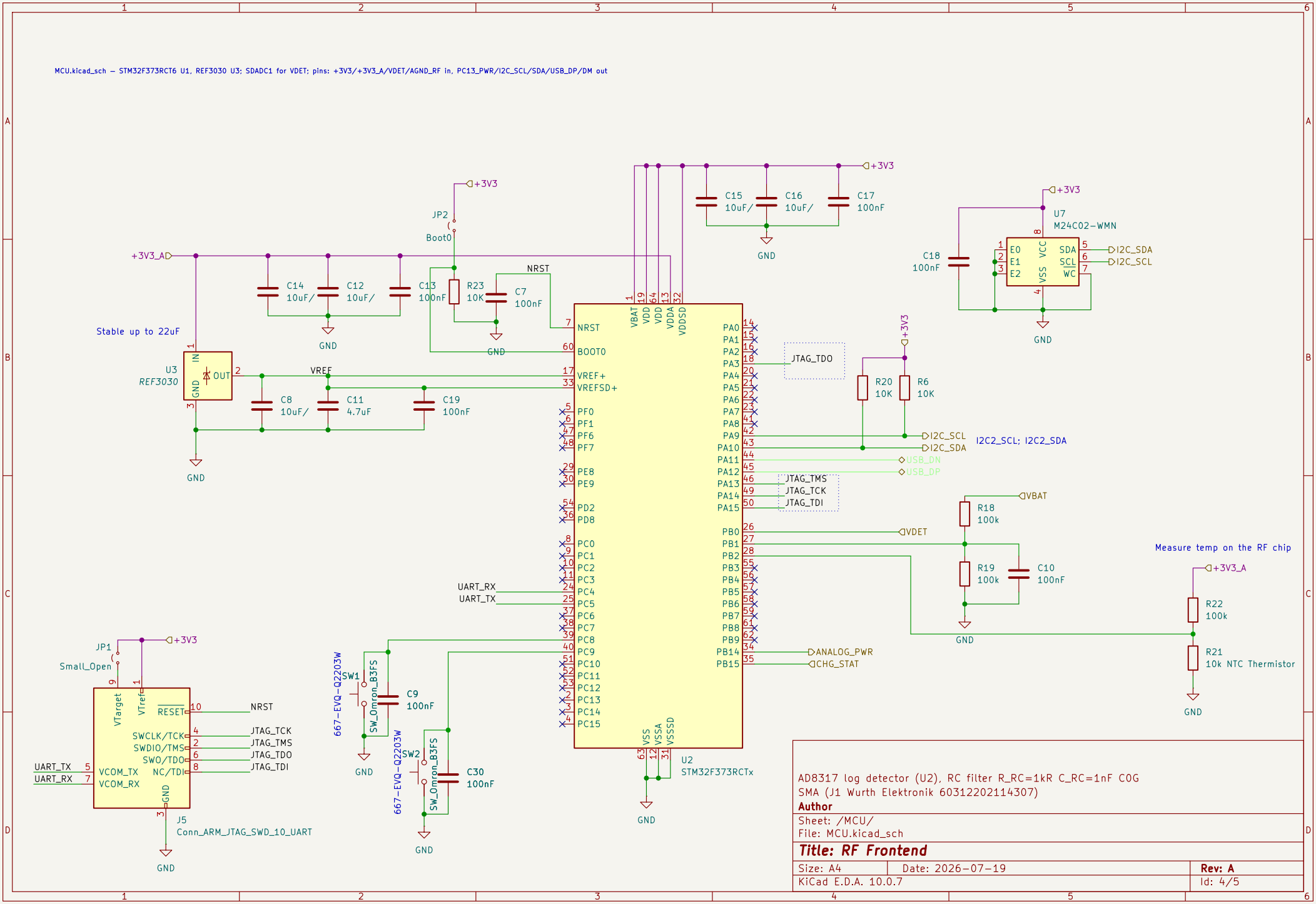
AP9101 protection (U7), FS8205A cell FETs (T_PCM), FB1-FB3 600R@100MHz
AD8317 log detector (U2), RC filter R_RC=1kR C_RC=1nF C0G
SMA (J1 Wurth Elektronik 60312202114307)

Author

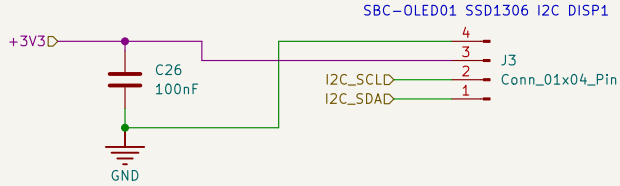
Sheet: /Power_Management/
File: Power_Management.kicad_sch

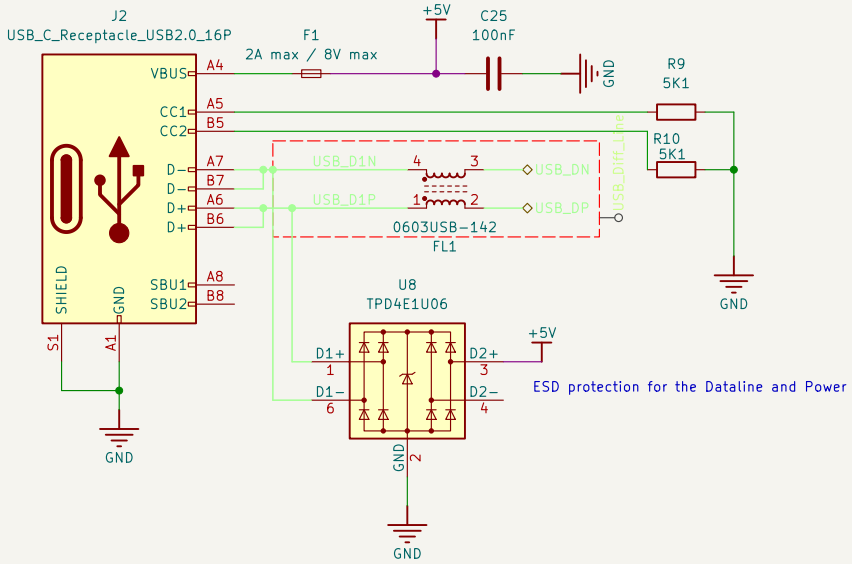
Title: RF Frontend

Size: A4	Date: 2026-07-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 3/5


MCU.kicad_sch — STM32F373RCT6 U1, REF3030 U3; SDADC1 for VDET; pins: +3V3/+3V3_A/VDET/AGND_RF in, PC13_PWR/I2C_SCL/SDA/USB_DP/DM out
+3V3
+3V3_A
JP2
Boot0
C15 10uF/
C16 10uF/
C17 100nF
GND
C14 10uF/
C12 10uF/
C13 100nF
R23 10K
C7 100nF
NRST
Stable up to 22uF
U3
REF3030
IN
OUT
GND
VREF
C8 10uF/
C11 4.7uF
C19 100nF
U7
M24C02-WMN
C18 100nF
E0
E1
E2
VCC
VSS
SDA
SCL
WC
I2C_SDA
I2C_SCL
VBAT
VDD
VDDA
VDDSD
NRST
BOOT0
VREF+
VREFSD+
PF0
PF1
PF6
PF7
PE8
PE9
PD2
PD8
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14
PC15
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB14
PB15
JTAG_TDO
R20 10K
R6 10K
I2C_SCL
I2C_SDA
I2C2_SCL; I2C2_SDA
USB_DN
USB_DP
JTAG_TMS
JTAG_TCK
JTAG_TDI
VDET
R18 100k
R19 100k
C10 100nF
ANALOG_PWR
CHG_STAT
Measure temp on the RF chip
R22 100k
R21 10k NTC Thermistor
UART_RX
UART_TX
JP1
Small_Open
VTarget
VTref
RESET
SWCLK/TCK
SWDIO/TMS
SWO/TDO
NC/TDI
VCOM_TX
VCOM_RX
J5
Conn_ARM_JTAG_SWD_10_UART
SW1
SW2
667-EVQ-Q2203W
SW_Omron_B3FS
C9 100nF
C30 100nF
VSS
VSSA
VSSSD
U2
STM32F373RCTx
AD8317 log detector (U2), RC filter R_RC=1kR C_RC=1nF C0G
SMA (J1 Wurth Elektronik 60312202114307)
Author
Sheet: /MCU/
File: MCU.kicad_sch
Title: RF Frontend
Size: A4
Date: 2026-07-19
Rev: A
KiCad E.D.A. 10.0.7
Id: 4/5

User_Interface.kicad_sch — SBC-OLED01 SSD1306 I2C DISP1, SW1 MENU PA0, SW2 NEXT PA1, USB-C CONN1 USB4110-GF-A; pins: I2C/+3V3/USB in, SW_MENU/SW_NEXT out





https://www.we-online.com/components/media/o109030v410%20ANP002d_The%20protection%20of%20USB%202.0%20applications.pdf

USB-C connector USB4110-GF-A (CONN1) — shared net with Power sheet
AD8317 log detector (U2), RC filter R_RC=1kR C_RC=1nF C0G
SMA (J1 Wurth Elektronik 60312202114307)

Author

Sheet: /User_Interface/
File: User_Interface.kicad_sch

Title: RF Frontend

Size: A4	Date: 2026-07-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 5/5